Student Name / Number : Vasukumar Patel [041153940]

```
> print(result)
  i   n Exact Trapezoid       relErr Simpson's       relErr
1 1   2   0.4 0.4267767 6.694174e-02        NA           NA
2 2   4   0.4 0.4070181 1.754528e-02 0.4004319 1.079790e-03
3 3   8   0.4 0.4018125 4.531162e-03 0.4000772 1.931236e-04
4 4  16   0.4 0.4004634 1.158503e-03 0.4000137 3.428367e-05
5 5  32   0.4 0.4001177 2.941780e-04 0.4000024 6.069614e-06
6 6  64   0.4 0.4000297 7.434966e-05 0.4000004 1.073534e-06
7 7 128   0.4 0.4000075 1.872977e-05 0.4000001 1.898112e-07
8 8 256   0.4 0.4000019 4.707610e-06 0.4000000 3.355641e-08
9 9 512   0.4 0.4000005 1.181352e-06 0.4000000 5.932130e-09
>
```

```r
f <- function(x) { x^(3/2) }

exact_value <- 2/5

student.trapezoid <- function(x0, xn, n){

  h <- (xn - x0) / n
  x <- seq(x0, xn, by = h)
  y <- f(x)

  approx <- (h / 2) * (y[1] + 2 * sum( y[2:n]) + y[n+1] )
  return(approx)

}

student.simpsons <- function(x0, xn, n){

  if(n == 2 || n %% 2 != 0){
    return(NA)
  }

  h <- (xn - x0) / n
  x <- seq(x0, xn, by = h)
  y <- f(x)

  odd_terms <- sum(y[seq(2, n ,by = 2)])
  even_terms <- sum(y[seq(3, n - 1, by = 2)])

  approx <- (h / 3) * (y[1] + y[n + 1] + 4 * odd_terms + 2 * even_terms)
  return(approx)

}

rel_error <- function(approx, exact) {
  abs(approx - exact) / abs(exact)
}

n_values <- 2^(1:9)

```

Student Name / Number : Vasukumar Patel [041153940]

```r

result <- data.frame(
  i = 1:length(n_values),
  n = n_values,
  Exact = exact_value,
  Trapezoid = NA,
  relErr = NA,
  `Simpson's` = NA,
  relErr2 = NA,
  check.names = FALSE
)

for (i in seq_along(n_values)) {
  n <- n_values[i]

  trap <- student.trapezoid(0, 1, n)
  simp <- student.simpsons(0, 1, n)

  result$Trapezoid[i] <- trap
  result$relErr[i] <- rel_error(trap, exact_value)

  result$`Simpson's`[i] <- simp
  result$relErr2[i] <- if (is.na(simp)) NA else rel_error(simp, exact_value)
}

names(result)[7] <- "relErr"

print(result)
```

1. **Which method gets accurate faster?**
   Simpson's method gets accurate faster than the trapezoid method.

2. **What information did you use to make this assertion?**
   I compared the relative error values in the table. Simpson's rule always has a much smaller error than the trapezoid rule for the same value of n. This shows it becomes accurate more quickly.

3. **If you wanted the approximation accurate to the third decimal place, which method and iteration would you have stopped at and why?**
   I would use Simpson's method and stop at n=4. At this point, the value is already accurate to three decimal places. The error is very small, so no need to increase n further.